S1 Supplementary Methods

Model variants and comparators, the Noetic Atlas readout streams, the Stage-1 NDHI composite, and truth-known simulator recoverability audits

Article type: Supplement
Short title: S1 Methods - NDT

Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

Relation to the main article

This supplement expands three appendix pointers from the main article (`NoeticDiffusionTheory.typ`, Appendix): **Model variants and comparators**, **Noetic Atlas: geometric and topological readouts**, and **Truth-known simulator audits**. None of the content here is required to evaluate the Stage-1 Minimal-MNDM / functional RVC / MNPS predictions; it supports later model comparison, exploratory geometric readouts, and estimator-validity discipline.

S1.0 Model variants and comparators

The MNDM can be systematically simplified to test which components are necessary:

1. **Minimal-MNDM:** Fixed geometry, no meta-field, and only the core (m, d, e) landscape. This is the simplest falsifiable form and should be tested first.
2. **Fixed-geometry (FG-MNDM):** Set $\boldsymbol{G}_t \equiv \boldsymbol{I}$ (identity). This tests whether adaptive attention geometry is necessary or whether rhythm-only scheduling suffices.
3. **No-meta (NM-MNDM):** Set $\Phi \equiv 0$ and $\delta \equiv 0$. This tests whether meta-regulation is necessary beyond the (m, d, e) landscape.
4. **Continuous-rhythm (CR-MNDM):** Use a smooth $r(t)$ rather than event-locked pulses. This tests whether discrete denoising steps are necessary or whether continuous modulation suffices.
5. **Extended-geometry MNDM:** Allow $\boldsymbol{G}_t$ and the landscape weights $(\alpha, \beta, \gamma, \delta)$ to vary. This form is reserved for later analyses or for cases where Minimal-MNDM and structured nulls fail.

These variants support hierarchical model comparison. A variant should be accepted only when it improves held-out prediction, survives multiverse robustness checks, and explains signatures that simpler rhythm or precision models cannot reproduce.

S1.1 Noetic Atlas: geometric and topological readouts

The Noetic Atlas applies four complementary analysis streams to MNPS-embedded data, each revealing different aspects of denoising dynamics:

Stream 1: Manifold embedding and geodesic flow. Low-dimensional projection (for example, PCA, PHATE, Laplacian eigenmaps) reveals intrinsic dimensionality and trajectory structure (Shine et al., 2016). We compute global dimensionality (participation ratio, intrinsic-dimension estimates), geodesic distances between states, local curvature (tangent-space PCA), and flow velocity and directionality along principal manifold axes.

Stream 2: Topological Data Analysis (Mapper/persistent homology). TDA reveals the "shape" of explored state space without assuming linearity. Mapper constructs a graph where nodes are clusters of similar states and edges encode transitions (Saggar et al., 2022; Saggar & others, 2018; Zhang et al., 2021). We extract hub nodes (high betweenness), component fragmentation (number of disconnected subgraphs), and topological features (loops, voids) via persistent homology (Betti numbers) (Sizemore et al., 2019).

Stream 3: Recurrence Quantification Analysis (RQA). RQA characterizes how trajectories revisit previous states. From recurrence plots we extract recurrence rate (RR), determinism (DET), laminarity (LAM), and entropy of line-length distributions (structural complexity of revisits) (Lopes et al., 2021).

Stream 4: Metastability and criticality indices. These quantify proximity to the critical regime where denoising is maximally informative: metastability index (variance of synchrony over time), avalanche-like branching-ratio estimates, and DFA exponents capturing long-range temporal correlations.

Each stream provides complementary information: Embedding reveals trajectory paths, TDA reveals manifold structure, RQA reveals temporal organization, and metastability indices quantify proximity to criticality. Together, they form a comprehensive geometric portrait of denoising dynamics.

S1.2 Stage-1 composite summary (NDHI)

The Noetic Diffusion Health Index (NDHI) is retained here as a **Stage-1 composite summary** for visualization and exploratory cross-condition comparison. It aggregates manifold dimensionality, curvature smoothness, hub/connector structure, recurrence, and metastability with equal weights in the first report and mandatory sensitivity analysis. The index is not used as a clinical biomarker, diagnostic label, or consciousness score.

This distinction is central. NDHI summarizes one operational question only: how close a measured trajectory appears to a metastable, flexible, near-critical regime under a specified measurement contract. Its value depends on the chosen MNPS chart, feature preprocessing, normalization, and weighting assumptions. Therefore every NDHI report must be accompanied by its component scores, confidence intervals or resampling uncertainty, and weight-perturbation sensitivity. Learned-weight variants may be reported for predictive benchmarking but are exploratory and not part of the primary construct definition.

The name “health” is used in this manuscript in the narrow dynamical sense above and should not be read as a normative, medical, or person-level evaluation. If future validation does not show reliability, cross-dataset stability, and outcome relevance beyond simpler features, NDHI should be treated as a plotting convenience rather than a validated endpoint.

S1.3 Truth-known simulator audits: recoverability and reachability

Source model. The programme forks from Slow Branch, a simulator-first model of replay-dependent slow branch-level accessibility (M_b) for memory linking (Langell, 2026a). In a canonical two-trace motif, replay-dependent slow writing reproduces a joint profile of overlap-branch strengthening, linking gain, perturbation-sensitive loss, and targeted rescue that is not reproduced by nine tested simpler or alternative comparator classes, and this profile generalizes across tested branch counts and motif classes. This is a model-**discrimination** result inside the simulator; it does not establish a unique molecular or cytoskeletal memory code, and M_b must never be renamed to an empirical MNPS or NDT coordinate.

The Slow Branch follow-on simulator programme supplies a methodological, rather than biological, extension to NDT. In truth-known synthetic systems it asks which latent distinctions survive a specified observation map and estimator. The four recoverability levels separate association, mechanism discrimination, latent-order preservation, and response-operator identification. This is useful discipline for empirical MNJ work: an observed chart need not identify the transition operator that generated it.

For local linearized synthetic dynamics, the epoch operator and propagated uncertainty are

$$A_{\text{epoch}} = J_{H-1} \cdots J_0, \quad \Sigma_{h+1} = A_h \Sigma_h A_h^T + Q. \tag{1}$$

The corrected simulator ladder recovered aggregate reachability calibration under its frozen contract, but its directional desired-versus-false-axis Gate R failed. Closure-signature recovery also depended on a saturation-avoiding regime. A separate audit found that a lag-embedded observer’s local response-generating operator was structurally null on the tested reachable channel (Gate J-O / E037R), while a deliberately lag-1-sensitive positive control confirmed the pipeline can detect a nonzero response block when one is present—an important negative result that bounds how far any chart-based estimate of a local transition operator can be trusted without a comparable positive-control check. These are internal validated and rejected results, respectively, for a synthetic model; they do **not** validate empirical MNJ estimates, MNPS reachability cones, or a micro-to-macro neural mechanism. Full contracts and gate ledgers are retained in the companion handover material (Langell, 2026b).

Status ceiling

Model-variant comparisons, Atlas readouts, NDHI, and simulator audits are all **methodological and exploratory** relative to the Stage-1 claim set. None of them substitutes for the event-locked entropy, sleep-curriculum topology, or bounded clinical-geometry tests defined in the main article.

Bibliography

Langell, R. (2026b,). *Separable Levels of Observability in a Truth-Known Memory Simulator*. Zenodo. https://doi.org/10.5281/zenodo.21823910

Langell, R. (2026a,). *Slow Branch-Level Accessibility as a Structural Constraint on Memory Linking*. Zenodo. https://doi.org/10.5281/zenodo.20813821

Lopes, M. A., Zhang, J., Krzemiński, D., Hamandi, K., Chen, Q., Livi, L., & Masuda, N. (2021). Recurrence Quantification Analysis of Dynamic Brain Networks. *Arxiv Preprint Arxiv:2106.12345*.

Saggar, M., & others. (2018). Towards a new approach to reveal dynamical organization of the brain using topological data analysis. *Nature Communications*, *9*, 1399. https://doi.org/10.1038/s41467-018-03664-4

Saggar, M., Supekar, K., & Menon, V. (2022). Precision dynamical mapping using topological data analysis reveals a hub-like transition state at rest. *Nature Communications*, *13*, 3210. https://doi.org/10.1038/s41467-022-32381-2

Shine, J. M., Koyejo, O., Bell, P. T., Ehgoetz Martens, K., Shine, R., Sporns, O., Breakspear, M., & Poldrack, R. A. (2016). Human cognition involves the dynamic integration of neural activity and neuromodulatory systems. *Nature Neuroscience*, *19*(3), 390–398. https://doi.org/10.1038/nn.4349

Sizemore, A. E., Phillips-Cremins, J. E., Ghrist, R., & Bassett, D. S. (2019). The importance of the whole: Topological data analysis for the network neuroscientist. *Plos Computational Biology*, *15*(4), e1006793. https://doi.org/10.1371/journal.pcbi.1006793

Zhang, M., Chowdhury, S., & Saggar, M. (2021). Temporal Mapper: Transition networks in simulated and real neural dynamics. *Plos Computational Biology*, *17*(8), e1008999. https://doi.org/10.1371/journal.pcbi.1008999